



Birla Institute of Technology & Science, Pilani

Work-Integrated Learning Programmes Division

MTech in Artificial Intelligence and Machine Learning

S2_2025-2026, AIMLCZG557/AECLZG557

Assignment 2 – PS12 - Bayesian Network - [Weightage 13%]

Read through this entire document very carefully before you start!

Problem Statement

A smart transportation system is being developed to monitor road conditions and predict traffic accidents under uncertain conditions. The system models probabilistic relationships among weather conditions, traffic congestion, road accidents, and emergency alerts using a Bayesian Network.

The intelligent agent must reason under uncertainty and estimate the probability of traffic accidents when only partial traffic information is available.

Use Bayesian inference principles and conditional probability computations to determine the likelihood of traffic accidents under varying traffic observations.

Task A: PEAS Description

Explain the PEAS (Performance Measure, Environment, Actuators, Sensors) model for the intelligent traffic monitoring agent. The explanation must be specific to traffic monitoring and accident prediction and should not be a generic PEAS description.

Task B: Bayesian Network Implementation

Implement the Bayesian Network using Python and perform probabilistic inference for the following scenarios.

General modelling assumption

All variables in this assignment are binary variables with values True and False.

Use the stated conditional independence assumptions to construct the required joint probability distribution. The implementation must use a reusable inference method based on joint enumeration or equivalent exact inference, rather than only hard-coded formulas for each query.

Scenario 1 – Road Accident Prediction

Network Structure

Road Accident (A) directly influences

- Traffic Delay (D)
- Emergency Call (E)

Directed edges:

$A \rightarrow D$

$A \rightarrow E$

There is no direct edge between Traffic Delay and Emergency Call.

Therefore,

D and E are conditionally independent given Road Accident.

Since both share the hidden common cause Road Accident, they are not marginally independent.

- Use the factorization: $P(A, D, E) = P(A) \times P(D | A) \times P(E | A)$.

Variables	Given Probabilities	Required Computations
Road Accident (A) Traffic Delay (D) Emergency Call (E)	$P(A)=0.07$ $P(D A)=0.88$ $P(D \neg A)=0.25$ $P(E A)=0.82$ $P(E \neg A)=0.10$	Construct the complete joint probability table. Compute $P(A D)$. Compute $P(A E)$. Compute $P(A D \wedge E)$. Determine whether D and E are independent without evidence. Check whether D and E are conditionally independent given Road Accident.

Scenario 2 – Traffic Signal Failure Detection

A smart traffic control system continuously monitors whether a traffic signal has failed using two separate monitoring units.

Traffic Signal Failure (S) influences

Camera Alert (C)

Sensor Alert (R)

Directed edges:

$S \rightarrow C$

$S \rightarrow R$

Both alerts depend on the same hidden cause (Traffic Signal Failure). Therefore,

C and R are not marginally independent.

They are conditionally independent given Traffic Signal Failure.

Use the factorization: $P(S, C, R) = P(S) \times P(C | S) \times P(R | S)$.

Independence clarification

Camera Alert and Sensor Alert are two separate monitoring channels. They are not marginally independent because both are influenced by the same hidden cause, Traffic Signal Failure. They are conditionally independent given Traffic Signal Failure.

Variables	Probabilities	Required Computations
Traffic Signal Failure (S) Camera Alert (C) Sensor Alert (R)	$P(S)=0.05$ $P(C S)=0.90$ $P(C \neg S)=0.12$ $P(R S)=0.87$ $P(R \neg S)=0.08$	Compute $P(S C)$. Compute $P(S R)$. Compute $P(S C \wedge R)$. Analyze whether C and R are independent without evidence. Check whether C and R are conditionally independent given Traffic Signal Failure.

Task C: Comparative Analysis

Compare the Bayesian Network approach with a deterministic rule-based traffic monitoring approach. The analysis should discuss:

- Difference between probabilistic reasoning and fixed IF–THEN rule-based reasoning.
- How Bayesian Networks handle uncertain, incomplete, or overlapping symptoms.
- Impact of conditional dependencies on inference accuracy.
- Effect of evidence propagation on posterior probabilities.
- Role of conditional independence and d-separation in reducing computational complexity.

Sample Input

```

PS12
  SCENARIO_1_ROAD_ACCIDENT
P_A=0.07
P_D_given_A=0.88
P_D_given_notA=0.25
P_E_given_A=0.82
P_E_given_notA=0.10

SCENARIO_2_TRAFFIC_SIGNAL_FAILURE

P_S=0.05
P_C_given_S=0.90
P_C_given_notS=0.12
P_R_given_S=0.87
P_R_given_notS=0.08

```

Note that the input/output data shown here is only for understanding and testing, the actual file used for evaluation will be different.

Sample Output

Scenario 1 - Joint Probability Table

A D E Probability

T	T	T	0.0505
T	T	F	0.0111
T	F	T	0.0069
T	F	F	0.0015
F	T	T	0.0233
F	T	F	0.2093
F	F	T	0.0698
F	F	F	0.6278

Scenario 1: Road Accident Prediction

$P(A \mid D) = 0.2095$

$P(A \mid E) = 0.3816$

$P(A \mid D \text{ and } E) = 0.6848$

Traffic Delay and Emergency Call are not independent without evidence.

Traffic Delay and Emergency Call are conditionally independent given Road Accident.

Scenario 2: Traffic Signal Failure Detection

$P(S \mid C) = 0.2830$

$P(S \mid R) = 0.3640$

$P(S \mid C \text{ and } R) = 0.8111$

Camera Alert and Sensor Alert are not independent without evidence.

Camera Alert and Sensor Alert are conditionally independent given Traffic Signal Failure..

Sample test validation

The sample output is mathematically correct under the Bayesian Network structures stated above.

The sample represents a happy-path test case. Students should additionally test missing keys, duplicate keys, invalid probability values, malformed input lines, and invalid scenario identifiers.

Note that the input/output data shown here is only for understanding and testing, the actual file used for evaluation will be different.

Display the output in outputPSXX.txt.

Deliverables:

- PDF document **designPSXX <group id>.pdf** detailing your solution.
- **[Group id] _Contribution.xlsx** mentioning the contribution of each student in terms of percentage of work done. Columns must be “Student Registration Number”, “Name”, “Percentage of contribution out of 100%”. If a student did not contribute at all, it will be 0%, if all contributed then 100% for all.
- **inputPSXX.txt** file used for testing
- **outputPSXX.txt** file generated while testing
- .py file containing the python code. Do not fragment your code into multiple files
- Zip all of the above files including the design document in a folder with the name:
 - **[Group id] _A2_PSXX.zip** and submit the zipped file.
 - Group Id should be given as **Gxxx** where **xxx** is your group number. For example, if your group is 26, then you will enter **G026** as your group id.

Instructions:

- It is compulsory to make use of the data structure(s) / algorithms mentioned in the problem statement.
- Ensure that all data structure insert and delete operations throw appropriate messages when their capacity is empty or full. Also ensure basic error handling is implemented.
- For the purposes of testing, you may implement some functions to print the data structures or other test data. But all such functions must be commented before submission.
- Make sure that you read, understand, and follow all the instructions
- Ensure that the input, prompt and output file guidelines are adhered to. Deviations from the mentioned formats will not be entertained.
- The input, prompt and output samples shown here are only a representation of the syntax to be used. Actual files used to evaluate the submissions will be different. Hence, do not hard code any values into the code.
- Please note that the design document must include:
 - One alternate way of modeling the problem with the performance implications.
 - Writing a good technical report and well documented code is an art. Your report cannot exceed 4 pages. Your code must be modular and quite well documented.
 - You may ask queries [in this dedicated discussion section](#). Beware that only hints will be provided and queries asked **in other channels like MS Teams, emails, Taxila inbox will not be responded to.**

Deadline

- The strict deadline for submission of the assignment is **17th Aug 2026 11:55 PM IST**

(Monday).

- The deadline has been set considering extra days from the regular duration in order to accommodate any challenges you might face. No further extensions will be entertained.
- Late submissions will be evaluated as below. Submissions made between:
 - 17th Aug 2026 11:56PM to 18th Aug 2026 11:55PM → Deduct 2M for late submission and evaluation for 10 marks.
 - 18th Aug 2026 11:56PM to 19th June 2026 11:55PM → Deduct 6M for late submission and evaluation for 6marks.
 - Beyond 19th Aug 2026 11:55PM → No evaluations, 0 marks will be awarded.

How to submit

- This is a group assignment.
- Each group has to make one submission (only one, no resubmission) of solutions.
- Each group should zip all the deliverables in one zip file and name the zipped file as [Group id] _A2_PSXX.zip and submit the zipped file.
- Group Id should be given as Gxxx where xxx is your group number. For example, if your group is 26, then you will enter G026 as your group id.
- Assignments should be submitted via Taxila > Assignment section. Assignments submitted via other means like email etc. will not be graded.

Evaluation

- The assignment carries **13 Marks**.
- Grading will depend on:
- Fully executable code with all functionality working as expected
 - Well-structured and commented code
 - Accuracy of the design document.
 - Every bug in the functionality will have negative marking.
 - Marks will be deducted if your program fails to read the input file used for evaluation due to change / deviation from the required syntax.
- We encourage students to take the upcoming assignments and examinations seriously and submit only original work. Please note that plagiarism in assignments will be taken seriously. Refer to the detailed policy [here](#).
- Source code files which contain compilation errors will get at most 25% of the value of that question.

Readings

Artificial Intelligence: A Modern Approach Textbook by Peter Norvig and Stuart J. Russell, 4th Edition.